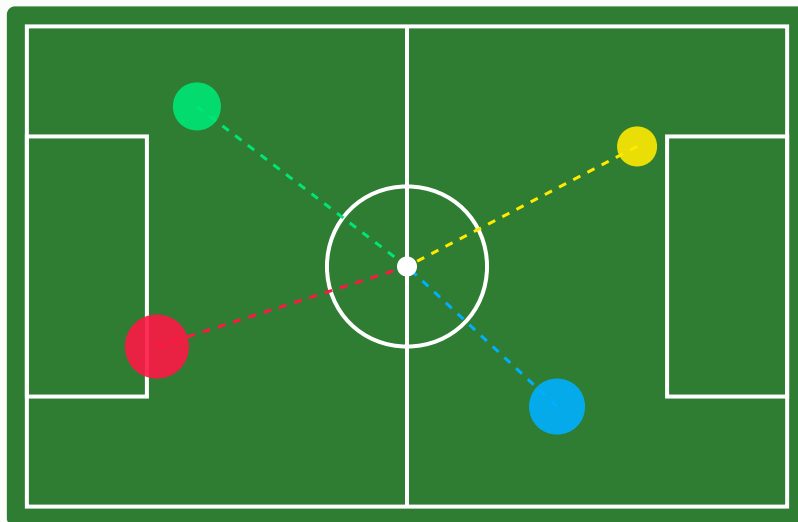


NEXUSPITCH

"Understand Every Moment of the Beautiful Game"

IBM June Innovation Challenge 2025 — Project Report



Team / Author: [Your Team Name]

Date: June 2025

The Problem

Over 3.5 billion fans watch football worldwide, but the reality is that the vast majority don't truly understand the complex, rapidly evolving situations happening on the pitch. When high-stakes decisions occur, confusion follows.

- VAR Confusion:** VAR decisions often take several minutes with zero explanation given to fans in the stadium or watching at home.
- Complex Rule Misunderstanding:** Niche scenarios regarding offsides, handball phases, and DOGSO (Denying an Obvious Goal-Scoring Opportunity) rules are misunderstood by the majority of viewers.
- Tactical Blindspots:** Crucial tactical substitutions, off-the-ball movements, and mid-game formation changes go largely unexplained during live matches.
- Lack of Real-Time Tools:** No accessible AI tool currently exists that can accurately analyze and explain football moments in real-time for fans.
- Accessibility Barriers:** Cultural, language, and football-literacy barriers make understanding even harder, especially at a World Cup scale.

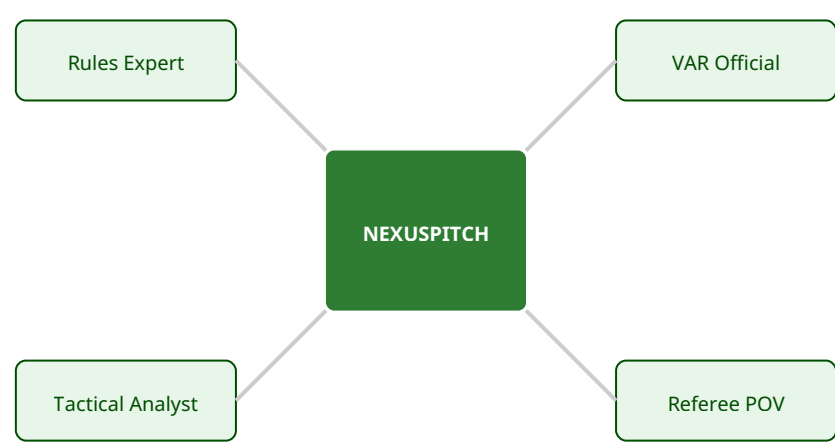


3.5B Viewers | 0 Real-Time Explanations | 1 AI Solution

The Solution — NEXUSPITCH

NEXUSPITCH is an intelligent conversational agent that demystifies football events in real-time. By leveraging large language models grounded in the official rules of the sport, NEXUSPITCH instantly analyzes scenarios and explains them through multiple expert perspectives, backed by dynamic visual animations.

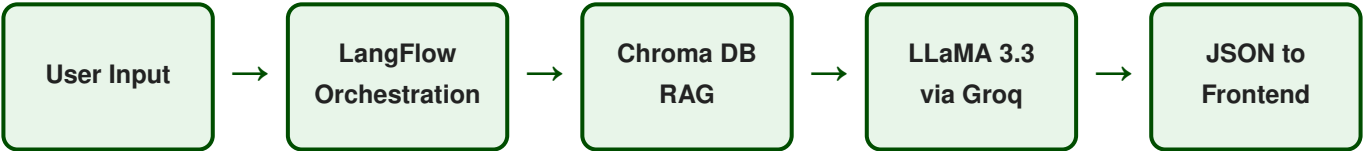
- Multi-Expert AI Breakdown:** Every query is answered through 5 lenses: Rules Expert, Tactical Analyst, VAR Official, Referee POV, and Overall Match Impact.
- Grounded RAG Pipeline:** Hallucinations are prevented by grounding responses strictly in the official FIFA Laws of the Game using a Retrieval-Augmented Generation approach.
- Dynamic 2D Pitch Animations:** Output is tied to dynamic HTML Canvas animations unique to every scenario to visualize the explanation.
- Conversation Memory:** The system supports robust follow-up and drill-down questions, allowing fans to probe deeper into why a call was made.
- Real Historic Context:** Responses automatically surface historic World Cup examples to bridge the gap between theory and real-world application.



Technical Architecture

NEXUSPITCH's architecture is designed for speed, accuracy, and rich frontend interactivity. The process flows smoothly from user input to visually rendered insights:

When a user types a scenario, it is ingested by the **LangFlow** orchestration engine. An intent classification node determines the context. The query is then routed to a **Chroma DB** RAG instance, which searches chunked embeddings of the official FIFA Laws of the Game. The context and prompt are passed to **LLaMA 3.3-70B** (served via the Groq API) to generate a highly structured JSON response. Finally, the frontend parses this JSON to populate expert panels and trigger exact Canvas API animations.



Component	Technology Used
Frontend	Pure HTML5, CSS3, JavaScript, Canvas API
AI Orchestration	LangFlow (IBM supported orchestration tool)
LLM / Inference	LLaMA 3.3-70B via Groq API
RAG Pipeline	Chroma DB + High-Speed Embeddings
Knowledge Base	Official FIFA Laws of the Game (Markdown)
Memory	LangFlow Message History (Store + Retriever)
Live Context	NewsAPI.org Integration

AI Response Schema

To enable the dynamic frontend to render animations and segregate expert opinions without additional parsing logic, NEXUSPITCH forces LLaMA to return data in a strict, predictable JSON schema. This guarantees that visual parameters are completely decoupled from the textual explanation.

```
{
  "response_type": "scenario",
  "text_response": "Overall summary of the event...",
  "incident_steps": ["Step 1", "Step 2", "Step 3"],
  "rules_expert": "Technical breakdown based on Laws...",
  "law_reference": "Law 12 - Fouls and Misconduct",
  "tactical_analyst": "How this affects the team's shape...",
  "referee_pov": "What the ref is looking at in real-time...",
  "var_official": "The protocol for reviewing this incident...",
  "real_example": "2018 World Cup: France vs Argentina...",
  "match_impact": "Red card impacts team by...",
  "dynamic_visuals": {
    "pitch_zone": "penalty_box",
    "ball_start": [x1, y1],
    "ball_end": [x2, y2],
    "attacker": [x, y],
    "defender": [x, y],
    "visual_caption": "Defender intercepts with arm"
  },
  "animation_type": "penalty_foul",
  "follow_up_prompt": "Would you like to know about...",
  "why_it_matters": "Understanding DOGSO is crucial..."
}
```

Why it matters: Structured outputs eliminate parsing errors and allow seamless mapping of coordinates (`ball\_start`, `pitch\_zone`) to HTML Canvas drawing functions.

Sample Interaction

Here is an example of NEXUSPITCH in action, handling a complex rule query:

User Query

"A defender deliberately handles the ball inside the penalty box to stop a clear goal scoring opportunity. What is the correct call and does VAR review this?"

AI Response Breakdown

Rules Expert The correct decision is a Penalty Kick and a Red Card for the defender. Under DOGSO guidelines, deliberately handling the ball to prevent a goal overrides the 'double jeopardy' yellow-card rule.

VAR Official Yes, VAR will review this automatically as it involves a direct Red Card incident and a Penalty Kick decision. Play is paused until clearance.

Tactical Analyst Losing a central defender forces an immediate substitution, typically sacrificing an attacker to bring on a substitute center-back, shifting the team to a low defensive block.

Historic Context Similar to the 2010 World Cup quarter-final (Uruguay vs Ghana) where Luis Suárez handled the ball on the line.

Triggered Visuals: animation_type: "penalty"

Law Reference: Law 12 (Fouls and Misconduct)

Why This Matters

As we approach the 2026 FIFA World Cup, billions of fans will tune in from diverse backgrounds. The sport is universal, but its nuances are often gatekept by technical jargon and confusing officiating.

Explainability as a Core AI Value: NEXUSPITCH brings transparency to chaotic moments, embodying IBM's ethos of trustworthy and explainable AI.

Bridging Knowledge Gaps: By breaking down events across different personas (Tactical, Rules, VAR), fans can learn at their own pace and interest level.

Cross-Cultural Understanding: Future iterations of NEXUSPITCH aim to integrate real-time translation, ensuring football remains accessible across all languages.

Future Scope & Scalability: The architecture is primed for integrating Live Match Data feeds (e.g., Opta stats), mobile app deployment, and voice narration features for visually impaired fans.

Conclusion

NEXUSPITCH is more than just an AI chatbot; it is a holistic educational tool designed to elevate how fans experience and understand football. By combining large language models, retrieval-augmented generation, and dynamic interactive visuals, we are empowering viewers to move from passive spectators to informed enthusiasts.

Tech Stack Summary: LangFlow + LLaMA 3.3 + Chroma DB + HTML5 Canvas

GitHub Repository: github.com/yourusername/nexuspitch

NEXUSPITCH — Where AI Meets the Beautiful Game.